USER MANUAL

3-Axis Digital Robotic Arm Controller ASIC

`tt_um_brazo_digital`

Version 1.0 | TinyTapeout Platform | 50 MHz

1. Overview

This document provides a comprehensive guide for using the tt_um_brazo_digital ASIC — a custom hardware chip designed to control a 3-Axis Digital Robotic Arm. The controller can drive two NEMA 17 stepper motors (one for the X-Axis and one for the Y-Axis) using standard push-buttons, and also includes a smart closed-loop interface for an automated gripper motor.

One of the key strengths of this chip is its built-in signal conditioning circuitry. Every physical button input is filtered through hardware-level synchronizers and 12-bit digital debouncers, which means mechanical button bounce and electrical noise are eliminated before they ever affect the motor outputs. This makes the controller robust and reliable in real-world environments.

2. Technical Specifications

Parameter	Value
Target Platform	TinyTapeout ASIC
System Clock	50 MHz
Reset Type	Asynchronous, Active-Low (rst_n)
Input Filtering	Dual flip-flop synchronizer + 12-bit digital debouncer per pin

Compatible Drivers	DRV8825, A4988, or any standard STEP/DIR driver

3. Pin Mapping

3.1 Input Pins (ui_in)

These pins accept signals from physical push-buttons and switches. All inputs are internally debounced, so you do not need external filtering circuits — a simple 10 kΩ pull-down resistor per pin is sufficient.

Pin	Signal	Function
ui_in[0]	noisy_btn_x_cw	Move X-Axis to the Right (Clockwise)
ui_in[1]	noisy_btn_x_ccw	Move X-Axis to the Left (Counter-Clockwise)
ui_in[2]	noisy_btn_y_cw	Move Y-Axis Up (Clockwise)
ui_in[3]	noisy_btn_y_ccw	Move Y-Axis Down (Counter-Clockwise)
ui_in[4]	noisy_dip_grip	Gripper Toggle — activate to open or close the gripper
ui_in[5]	noisy_sw_limit	Force Limit — gripper stops automatically when this is triggered
ui_in[6]	(Unused)	Leave floating or connect to GND
ui_in[7]	(Unused)	Leave floating or connect to GND

3.2 Output Pins (uo_out)

These pins generate STEP and DIR signals to be connected directly to stepper motor driver boards such as the DRV8825 or A4988.

Pin	Signal	Target Driver Pin	Description
uo_out[0]	step_x	STEP (X Driver)	Step pulses for X-Axis motor
uo_out[1]	dir_x	DIR (X Driver)	Direction signal for X-Axis
uo_out[2]	step_y	STEP (Y Driver)	Step pulses for Y-Axis motor
uo_out[3]	dir_y	DIR (Y Driver)	Direction signal for Y-Axis
uo_out[4]	step_grip	STEP (Gripper Driver)	Step pulses for Gripper motor
uo_out[5]	dir_grip	DIR (Gripper Driver)	Direction signal for Gripper
uo_out[6]	—	None	Permanently LOW (unused)
uo_out[7]	—	None	Permanently LOW (unused)

3.3 Bidirectional Pins (uio)

All bidirectional pins (uio[7:0]) are configured as high-impedance inputs and are not used by this design. You can leave them unconnected or tied to GND.

4. How to Use the Controller

4.1 Startup and Initialization

Follow these steps every time you power on the system:

- Connect a stable 50 MHz clock signal to the clk pin.
- Pull the rst_n pin LOW (GND) for at least 3 clock cycles to reset the controller.
- Release rst_n by connecting it to VCC. The chip will enter its idle state: all step signals go LOW, all counters reset to zero.

Note: If you observe erratic behavior at startup, verify that the reset sequence is properly executed and that the clock source is stable before releasing the reset line.

4.2 X-Axis and Y-Axis Movement

The X and Y axes are controlled by press-and-hold push-buttons. The controller responds as long as the button is held down and stops immediately upon release.

Action	Button	DIR Signal	Result
X-Axis Right	ui_in[0]	dir_x = 1	Motor steps in CW direction
X-Axis Left	ui_in[1]	dir_x = 0	Motor steps in CCW direction
Y-Axis Up	ui_in[2]	dir_y = 1	Motor steps in CW direction
Y-Axis Down	ui_in[3]	dir_y = 0	Motor steps in CCW direction

Safety Interlock: If both the CW and CCW buttons for the same axis are pressed at the same time, the controller automatically stops the motor to prevent damage. The step signal halts until only one button remains pressed.

4.3 Gripper Control

The gripper uses a toggle-based control system combined with a force feedback input for safe object handling:

- Triggering the Gripper: Activate ui_in[4] (the DIP switch or latch input) to toggle the gripper between its open and closed states.
- Force Limit Protection: While the gripper is closing, if ui_in[5] becomes active (e.g., a contact switch detects resistance), the step_grip signal stops immediately. This prevents motor stall and protects both the object and the gripper mechanism.

Note: The force limit input (ui_in[5]) acts as a digital end-stop. Make sure your hardware sensor provides a clean HIGH signal when contact is made.

5. Wiring and Circuit Integration

5.1 Typical Connection Diagram

The following describes the recommended wiring architecture for integrating the ASIC into a PCB or breadboard prototype:

External Component	Connects To	Notes
50 MHz Oscillator	clk pin	Must be stable before releasing reset
Reset Button (N.O.)	rst_n pin	Pull LOW momentarily, then release to HIGH
Push-Button X Right	ui_in[0]	Add 10 kΩ pull-down resistor to GND
Push-Button X Left	ui_in[1]	Add 10 kΩ pull-down resistor to GND
Push-Button Y Up	ui_in[2]	Add 10 kΩ pull-down resistor to GND
Push-Button Y Down	ui_in[3]	Add 10 kΩ pull-down resistor to GND
DIP Switch (Grip)	ui_in[4]	Toggle switch to activate gripper
Limit Switch (Force)	ui_in[5]	End-stop sensor for gripper force feedback
DRV8825 (X-Axis)	uo_out[0], uo_out[1]	STEP and DIR for X motor
DRV8825 (Y-Axis)	uo_out[2], uo_out[3]	STEP and DIR for Y motor
DRV8825 (Gripper)	uo_out[4], uo_out[5]	STEP and DIR for Gripper motor

5.2 Hardware Recommendations

- Pull-Down Resistors: Each button input (ui_in[0] through ui_in[5]) should include a 10 kΩ resistor connected between the pin and GND. This ensures a clean LOW reference when the button is not pressed.

- Shared Ground Plane: The GND of all stepper motor drivers (DRV8825 units) must be connected to the same common ground as the ASIC power supply. Floating grounds can cause false triggering on the STEP and DIR lines.
- Clock Source Stability: Use a dedicated crystal oscillator module for the 50 MHz clock. Unstable clocks can cause missed steps or unpredictable behavior.
- Driver Microstepping: Set the microstepping mode on your DRV8825 boards according to your desired motor resolution. The ASIC outputs a fixed-frequency step train, so resolution is entirely determined by the driver configuration.

6. Troubleshooting

Symptom	Possible Cause	Suggested Fix
Motor does not move when button is pressed	rst_n not released, or no clock signal	Verify clock is active and reset line is HIGH
Motor moves erratically	Missing pull-down resistors on button inputs	Add 10 kΩ pull-down resistors to each input pin
Both directions cancel each other	CW and CCW buttons pressed simultaneously	Release one button — the interlock is working correctly
Gripper does not stop on contact	Limit switch not wired to ui_in[5]	Check wiring and verify sensor outputs a clean HIGH on contact
No output on STEP/DIR lines	Motor driver not powered or wired incorrectly	Verify DRV8825 power supply and signal connections
Gripper moves in wrong direction	DIP switch toggle state may be inverted	Toggle ui_in[4] to cycle the gripper state